

Total No. of Questions : 8]

SEAT No. :

PD4675

[6404]-181

[Total No. of Pages : 2

B.E. (Information Technology)

AUGMENTED AND VIRTUAL REALITY

(2019 Pattern) (Semester - VIII) (414452B) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Explain Perception of colors in detail. **[9]**

b) Explain concept of Visual Rendering in detail. **[9]**

OR

Q2) a) Explain concept of Motion in Real and Virtual Worlds in detail. **[9]**

b) Explain Three categories of tracking in VR systems. **[9]**

Q3) a) How does Augmented Reality Work? Explain with suitable example. **[9]**

b) Describe any two applications of Augmented Reality. **[8]**

OR

Q4) a) Define Augmented Reality? Explain History of Augmented Reality. **[8]**

b) Explain Ingredients of an Augmented Reality Experience. **[9]**

P.T.O.

Q5) a) What is hybrid tracking? Explain with suitable examples. **[9]**

b) What is role of sensor in tracking? Explain in detail. **[9]**

OR

Q6) a) Describe different types of AR devices. **[9]**

b) Describe sensor fusion with example. **[9]**

Q7) a) What is natural-feature tracking? Describe with any one application. **[9]**

b) What is simultaneous localization and mapping in AR? **[8]**

OR

Q8) a) What are the major software components in augmented reality? **[9]**

b) What is marker tracking in AR? Explain in detail. **[8]**

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